Bahir Dar University
BSc. In Information System
Chapter 1: Introduction to machine learning

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Bahir Dar , Ethiopia
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What is machine learning ?

- Many definitions have been proposed for the term machine learning (ML).
- Machine learning is a branch of artificial intelligence, concerned with the design and development of algorithms that ***allow computers to evolve behaviours*** based on empirical data.
- Arthur Samuel (1959) defined machine learning as "a subfield of computer science that gives computers the ability to learn without being explicitly programmed."
  - It means that ***ML is able to perform a specified task without being directly told how to do it.***
  - *Example:* distinguish between spam and valid email messages. G*iven a set of manually labelled good and bad email examples, an algorithm can automatically learn a set of rules that distinguish them.*

What is machine learning ?

- A widely accepted formal definition by Tom Mitchell (1997)
  - **A computer program is said to learn from experience *E* with respect to some class of tasks *T* and performance measure *P*, if its performance at the tasks *T* , as measured by *P*, improves with the experiences.**
- In short
  - ***a set of computer programs that automatically learn from past experiences (examples or training corpus)***
  - ***Improve*** the ***performances*** of intelligent programs

- Example: According to this definition, we can reformulate the email problem as the task of identifying spam messages (***task T)*** using the data of previously labelled email messages (***experience E)*** through a machine learning algorithm with the goal of improving the future email spam labelling (***measure P)***

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- Example: Handwritten recognition problem
  - ***Task T*** **:** *Recognizing hand written characters*
  - ***Performance measure P*** **:** *Percent of characters correctly classified*
  - ***Training experience E*****:** A database of handwritten characters with given classifications

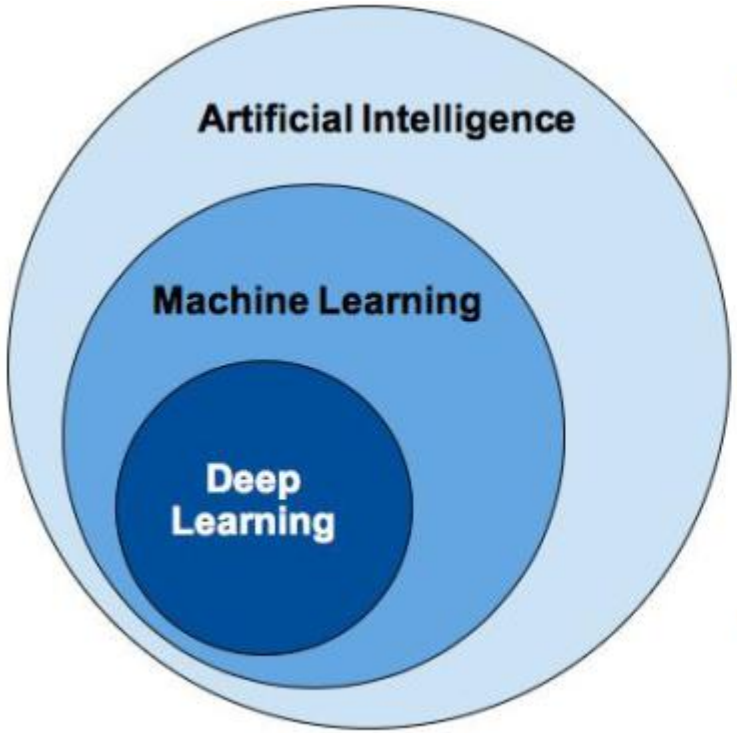
What is machine learning ?

- In the field of today's data science:

- ML aims to select, explore and extract useful knowledge from complex, often non-linear data, building a computational model capable of describing unknown patterns or correlations, and in turn, solve challenging problems.

- This learning process is often carried out through repeated exposure to the defined problem (dataset), allowing the model to achieve self-optimization and continuously enhance its ability to solve new, previously unseen problems.

Artitificial intelligence Vs. Machine Learning Vs. Deep learning



- **AI :** broader concept (science+Engineering) to create intelligent machines that can simulate human thinking capability.
- **ML :** subset of AI that allows machines to learn from data without being programmed explicitly.
- **DI :** subset of ML, that uses the neural networks to analyze different factors with a structure that is similar to the human neural system.

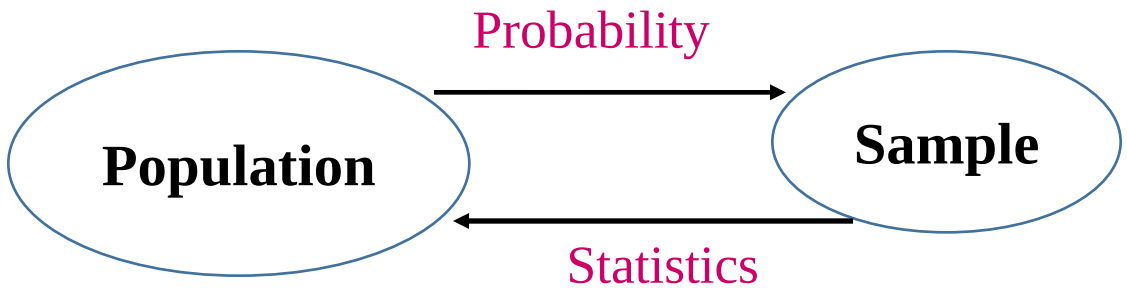
Probability Theory in Machine Learning

- **Probability** is key concept is dealing with uncertainty
- Three main sources of uncertainty in machine learning: **noisy data**, **incomplete coverage of the problem domain and imperfect models**.
- Probability Theory
  - Framework for quantification and manipulation of uncertainty
  - Provides the foundation and tools for quantifying, handling, and harnessing uncertainty in applied machine learning.
- Probability provides the basis for developing specific algorithms, such as Naive Bayes, and Bayesian Belief Networks

Basic Concepts on Probability

- **Probability**. The word *probability* is actually undefined, but the *probability of an event* can be explained as the proportion of times, under identical circumstances, that the event can be expected to occur.
  - It is the event's long-run frequency of occurrence.
  - For example, the probability of getting a head on a coin toss = .5. If you tossing a coin repeatedly, for a long time, you will note that a head occurs about one half of the time.

- **Probability vs. Statistics**.
  - In probability, the population is known.
  - In statistics you draw inferences about the population from the sample.

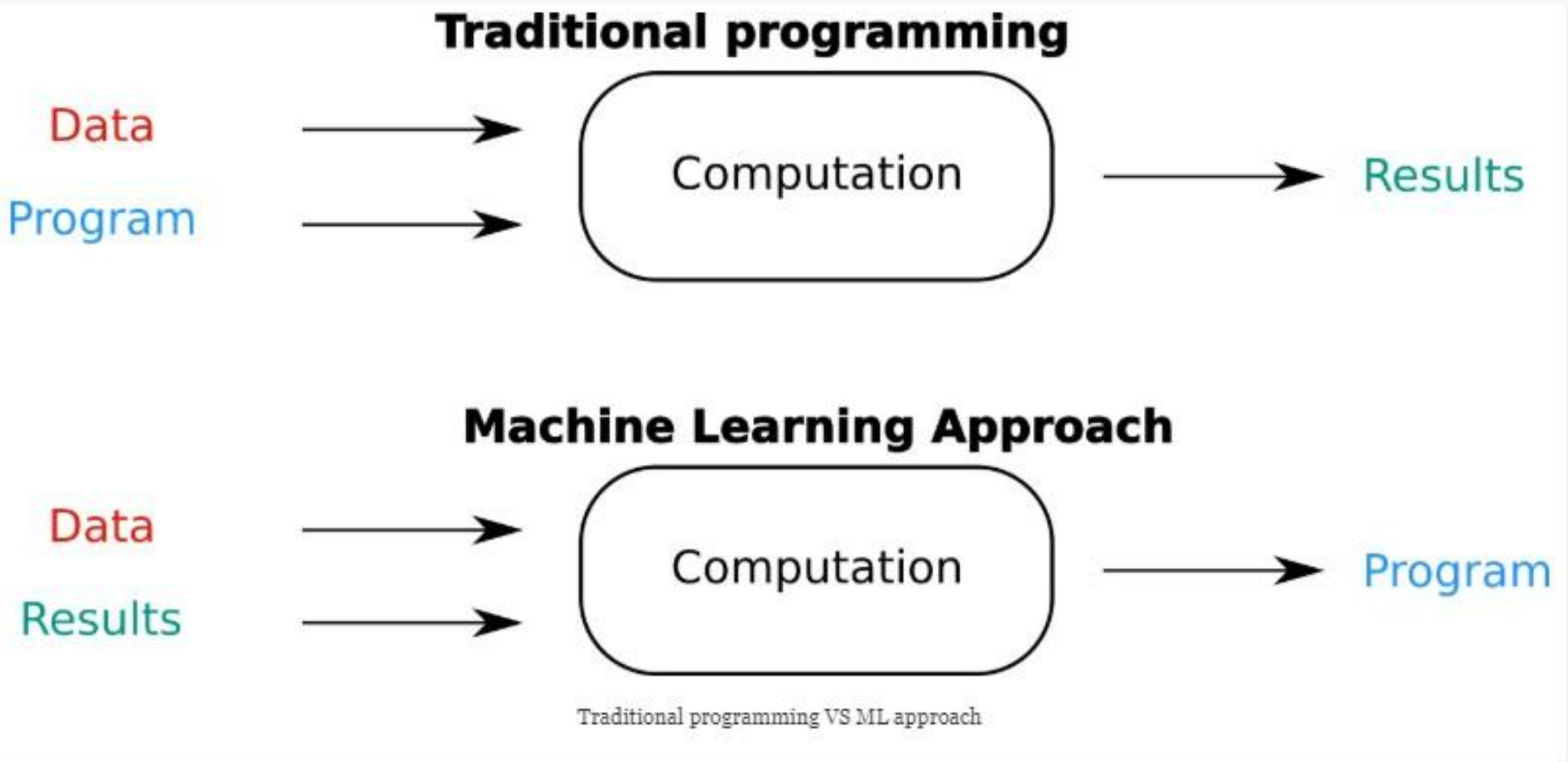


Probabilistic vs Statistical Reasoning

- Suppose **I know exactly** the proportions of car makes in California. Then I can find the probability that the first car I see in the street is a Ford. This is probabilistic reasoning as I know the population and predict the sample.

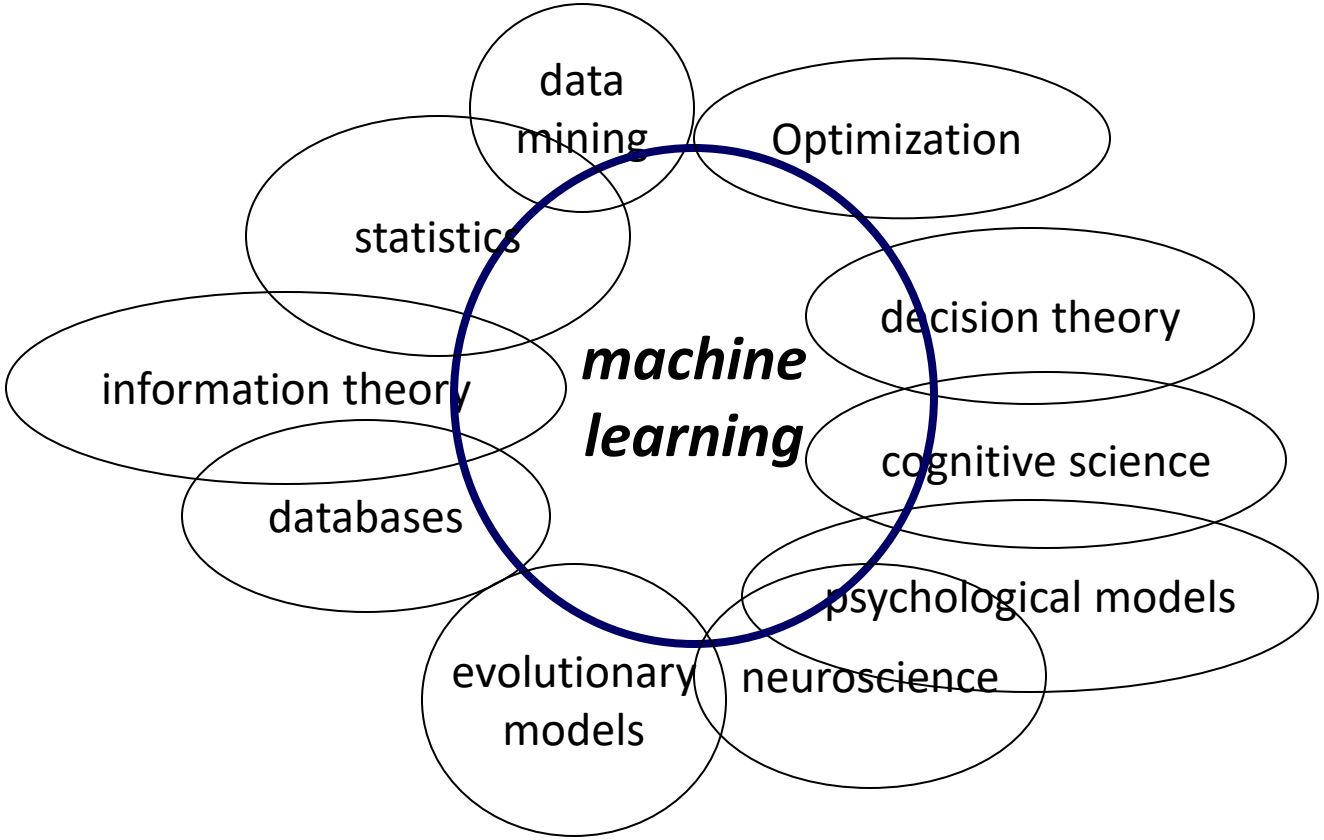
- Now suppose that **I do not know** the proportions of car makes in California, but would like to estimate them. I observe a random sample of cars in the street and then I have an estimate of the proportions of the population. This is statistical reasoning

Traditional vs ML Approach



Traditional programming VS ML approach

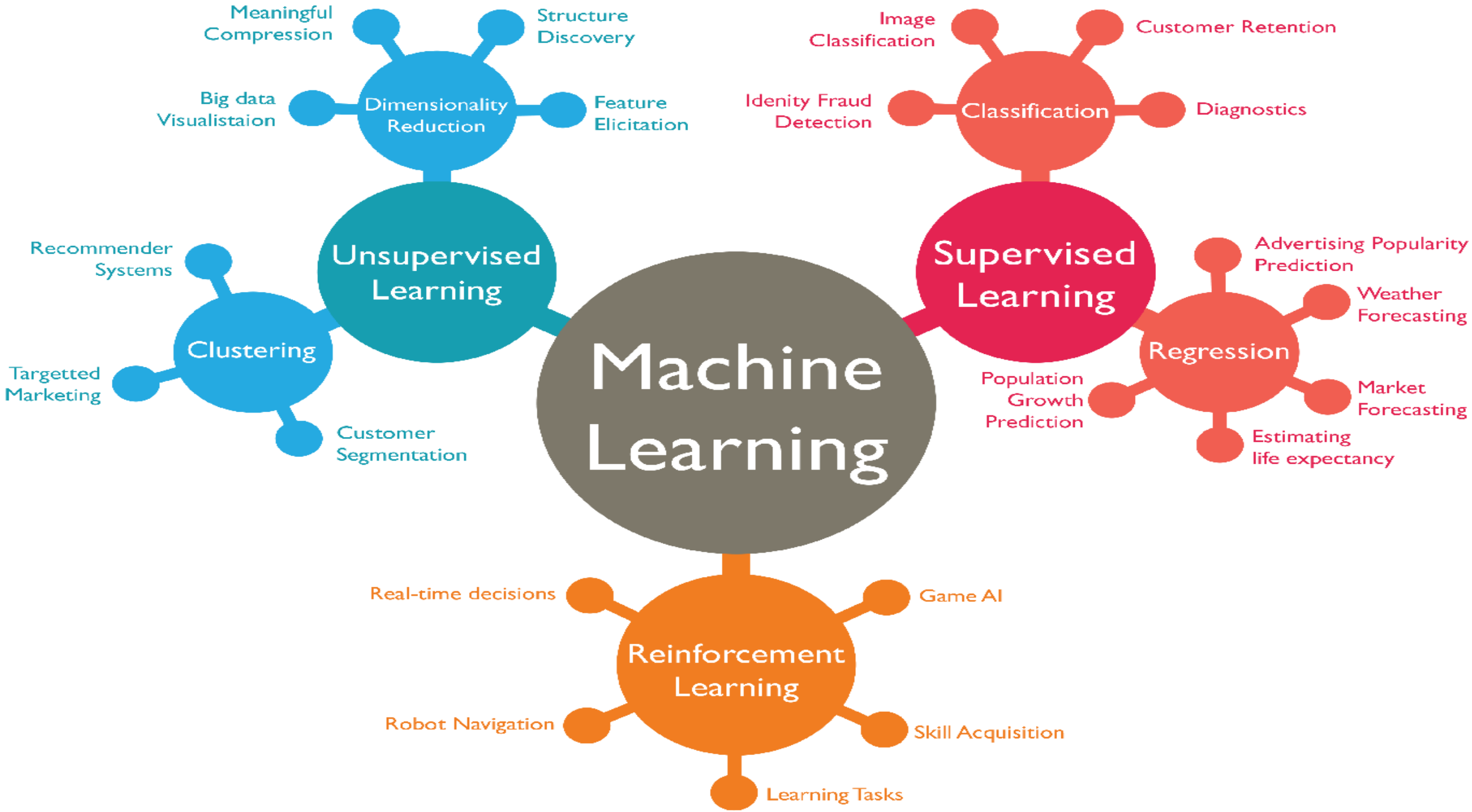
Related Fields



Machine learning is primarily concerned with the **accuracy** and **effectiveness** of the *computer system* in **performing complex tasks.**

Classes of Machine Learning problem

- **Supervised Learning**
  - Learn to predict output when given an input vector
  - Training data includes desired outputs
- **Unsupervised Learning**
  - The aim is to uncover the underlying structures (classes or clusters) in the data
  - Training data does not include desired outputs. This is the new frontier of machine learning because most big datasets do not come with labels.
- **Semi-supervised Learning**
  - Desired outputs or classes are available for only a part of the training data.
  - This approach is useful when it is impractical or too expensive to access or measure the target variable for all participants
- **Reinforcement Learning**
  - Learning method that interacts with its environment by **producing actions** and **discovers** errors or rewards.
  - On the basis of trial and error, to discover what actions maximize reward and minimize the penalty.



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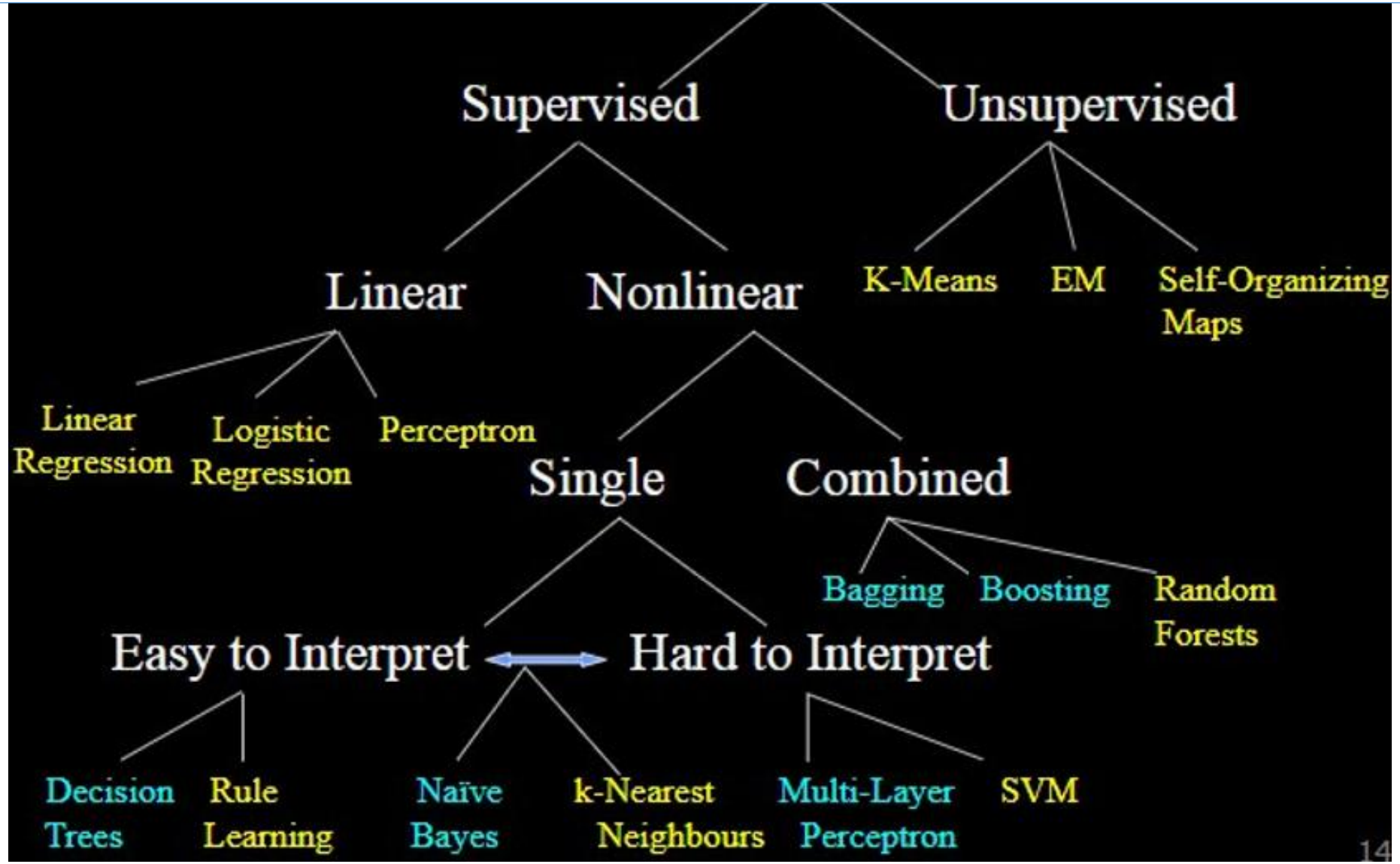
Classes of machine learning problem

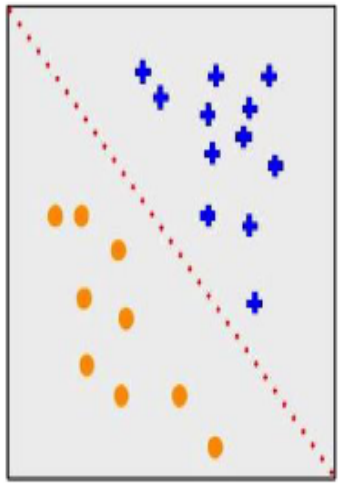


Classes of machine learning Problem

Supervised

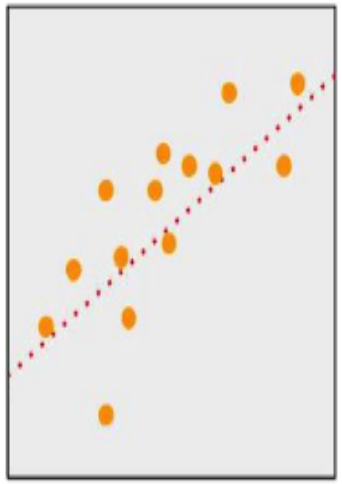
Classification

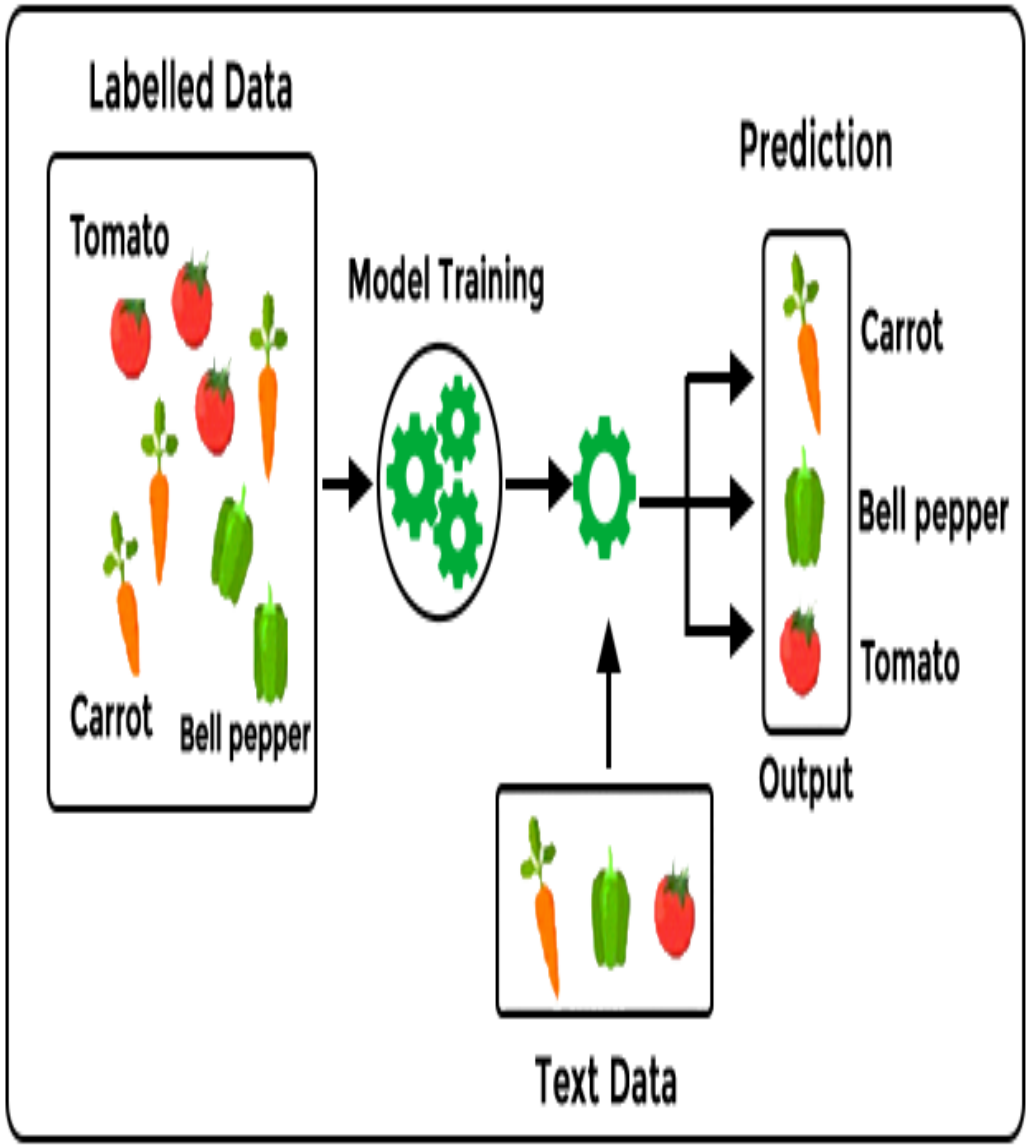
Output is a **discrete** variable (e.g., cat/dog)



Regression

Output is **continuous** (e.g., price, temperature)





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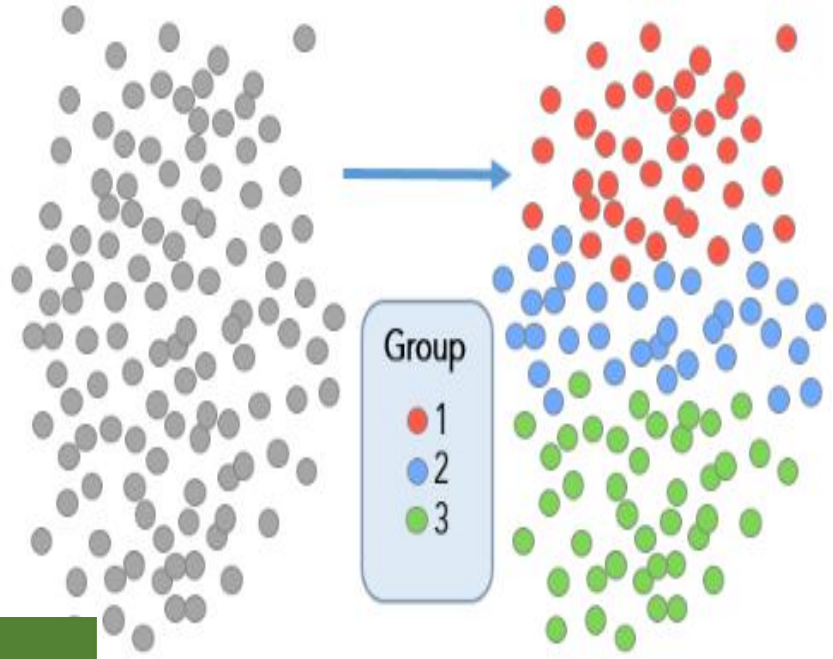
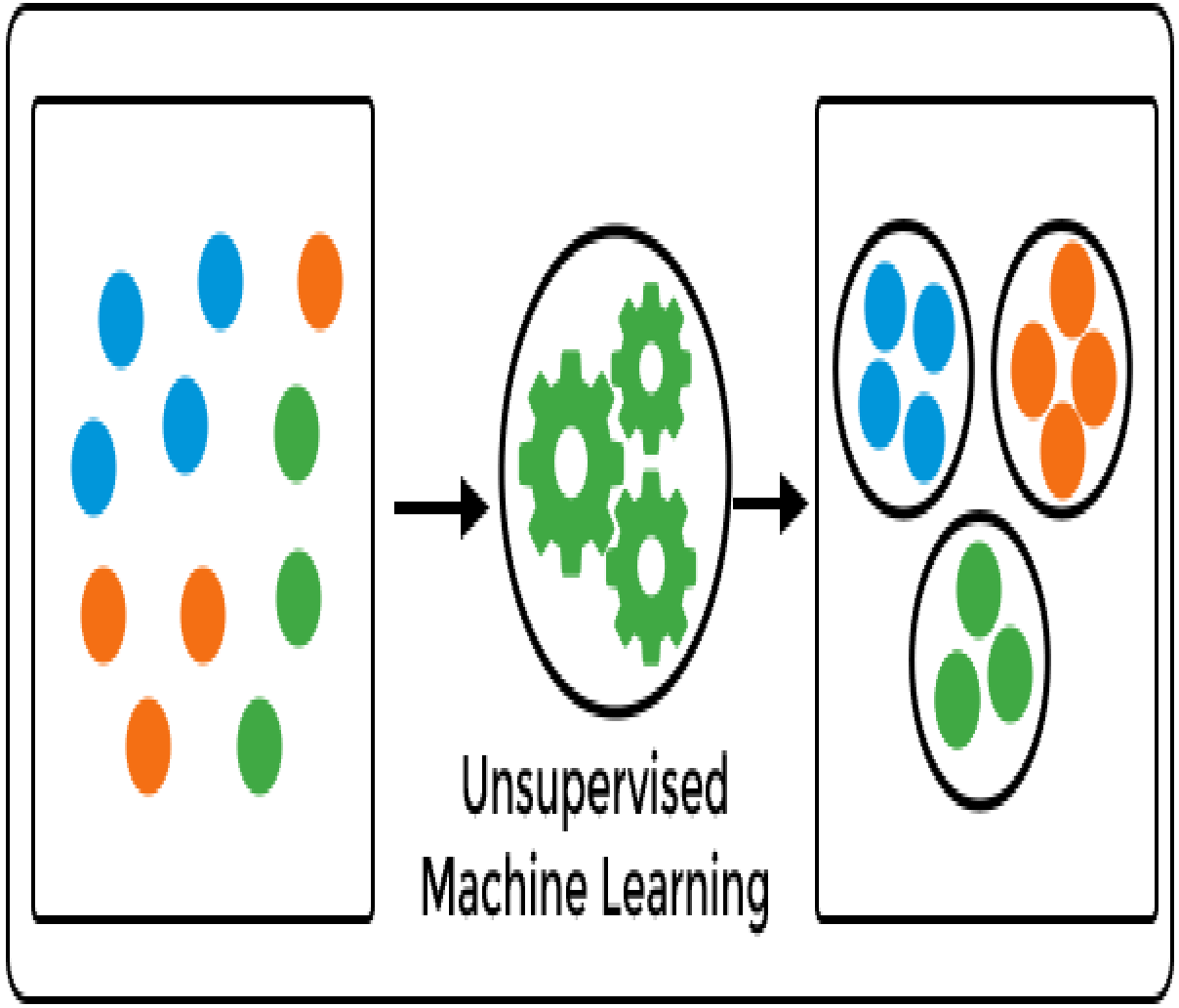
Classes of machine learning Problem

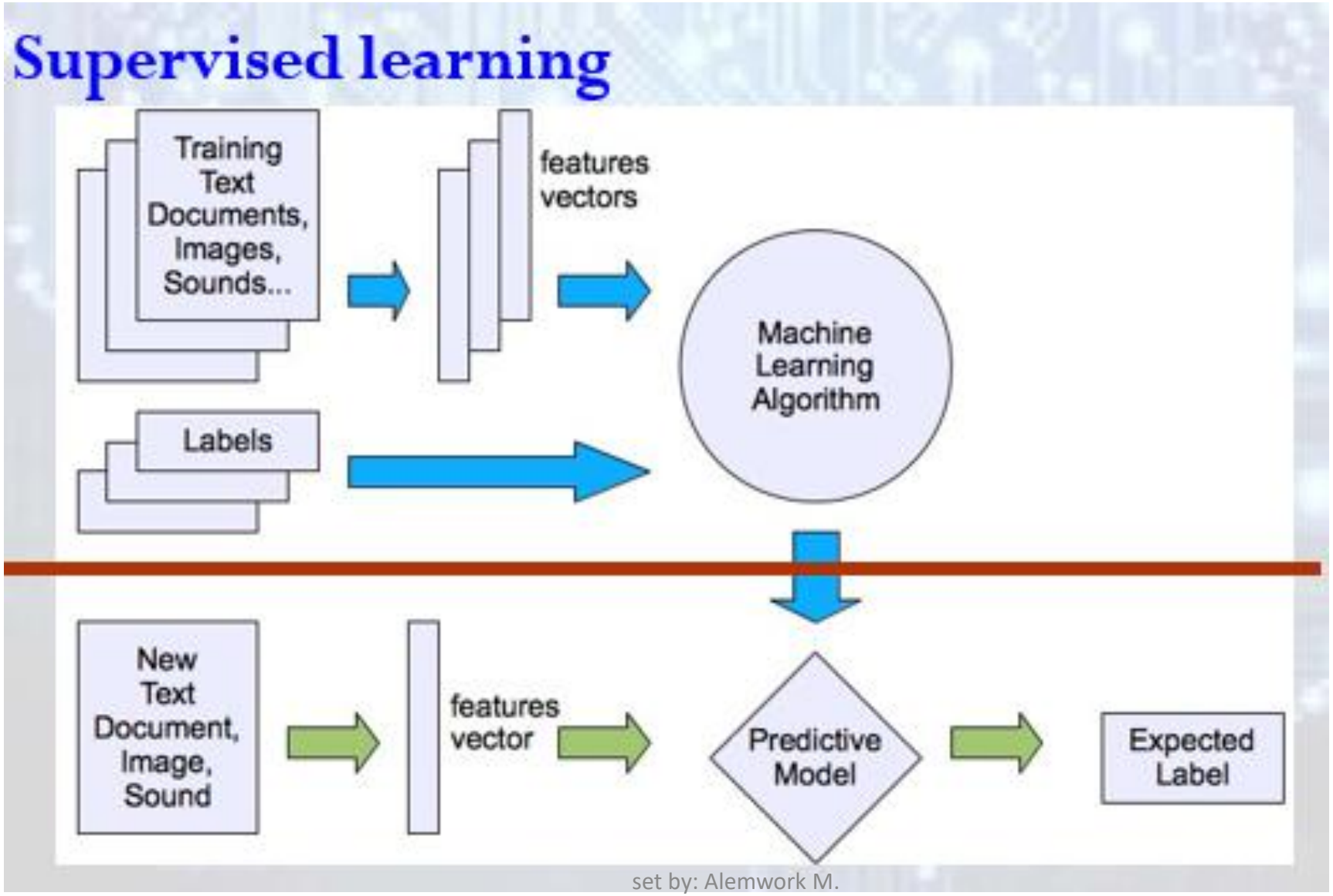
Unsupervised



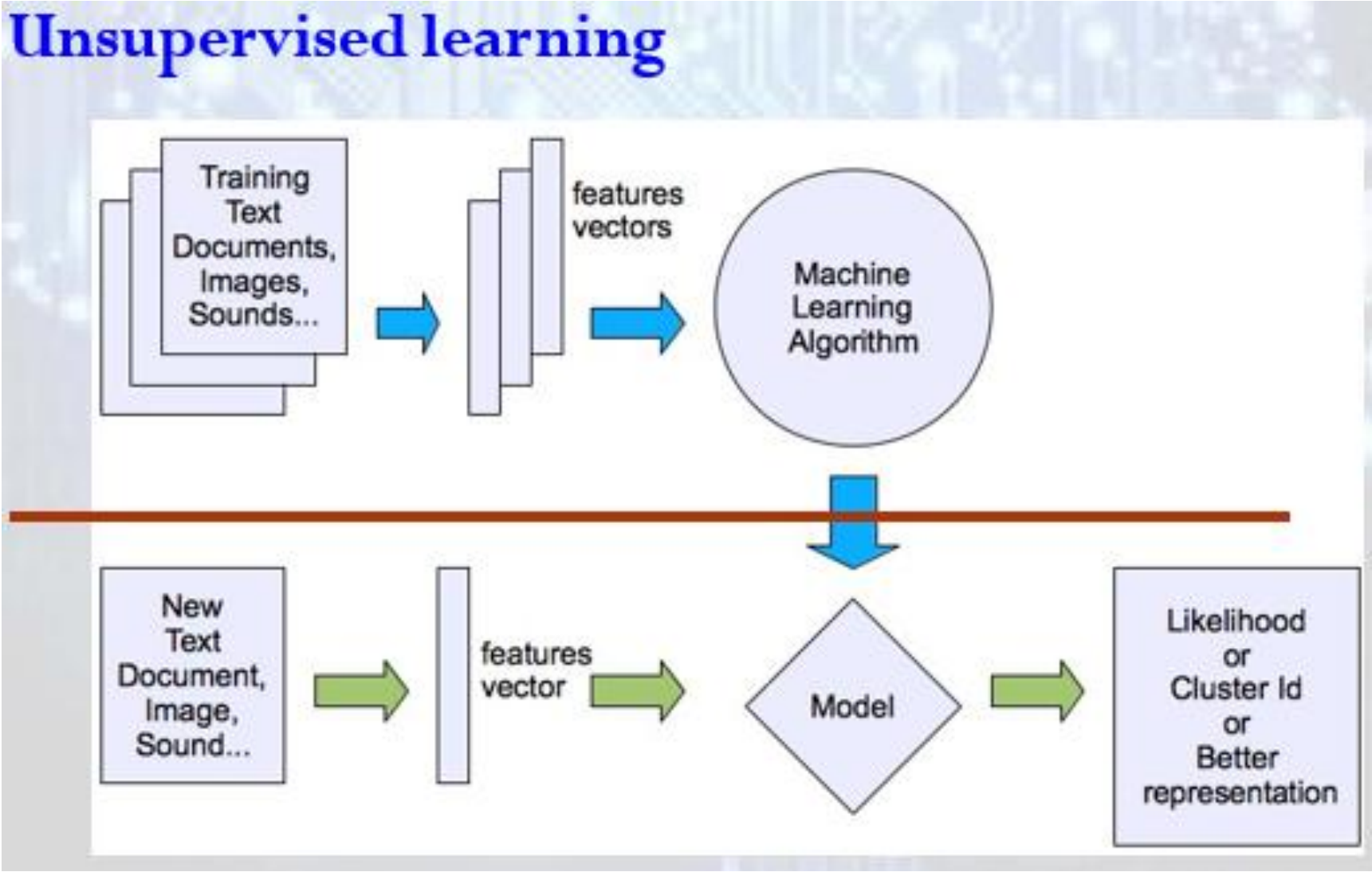
- Clustering
- Dimensionality reduction



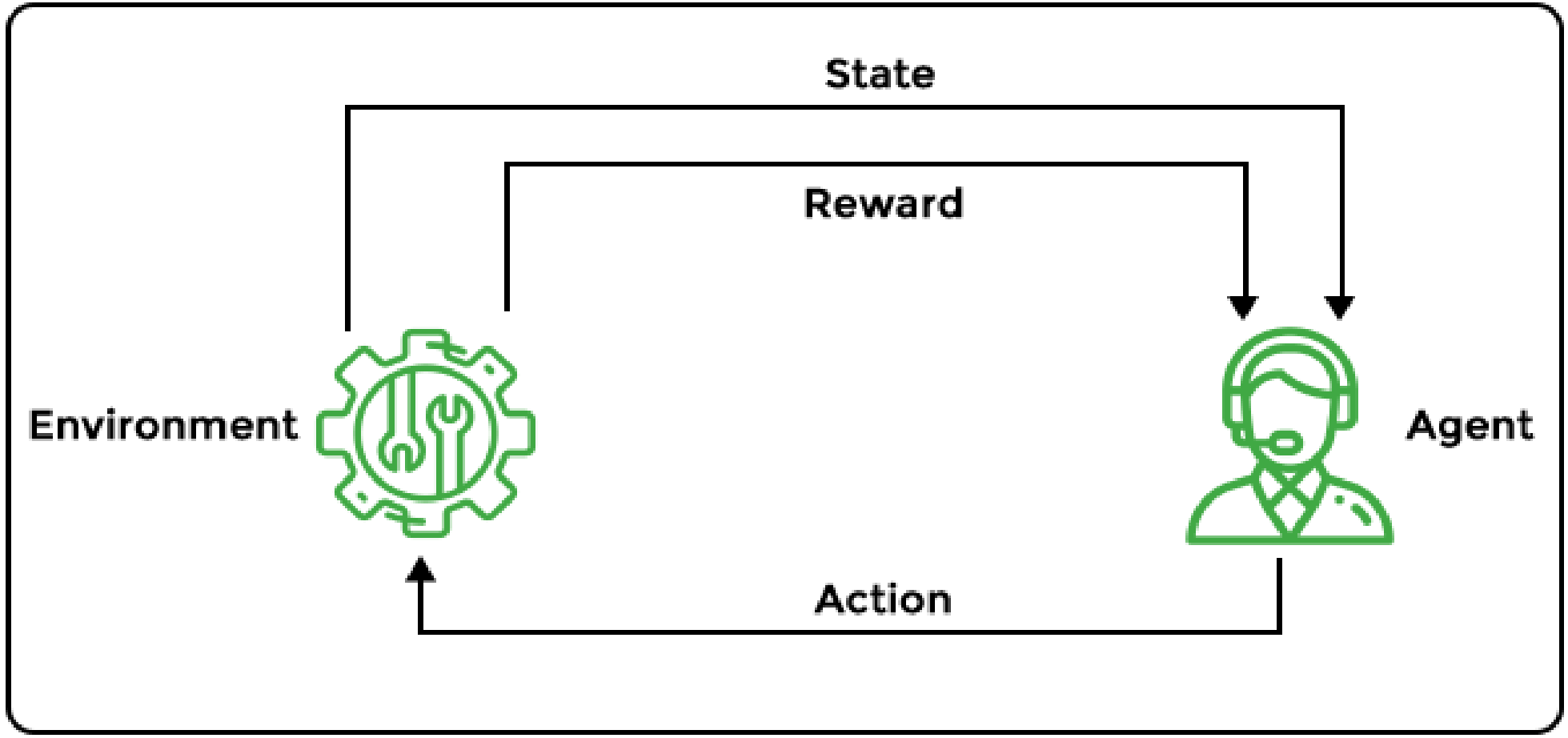
Machine learning structure



Machine learning structure



Reinforcement Learning



The machine learning framework

$$y = f(\mathbf{x})$$

output

prediction function

Image feature

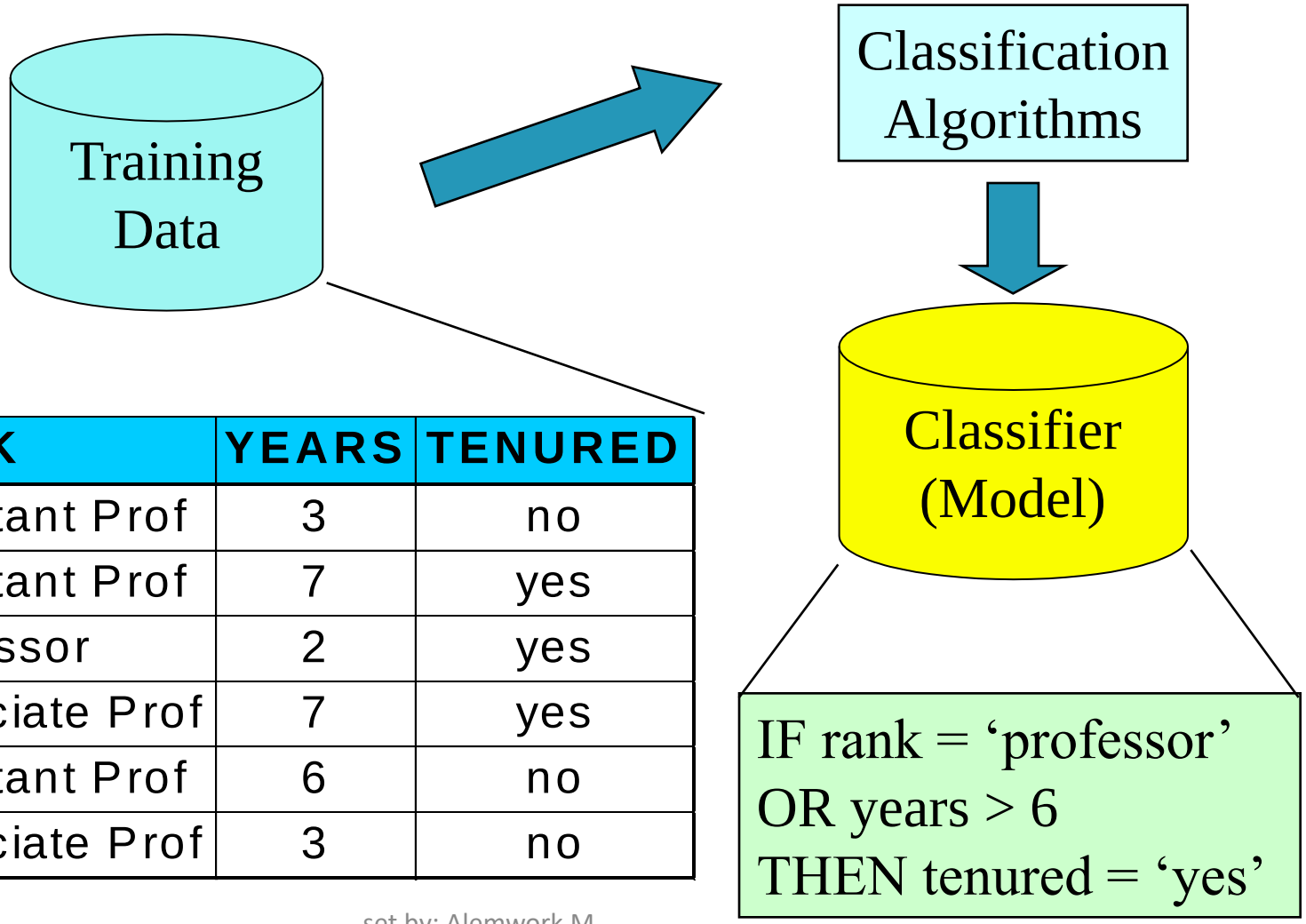
- **Training:** given a *training set* of labeled examples $\{(\mathbf{x}_1,y_1), \ldots, (\mathbf{x}_N,y_N)\}$, estimate the prediction function $\mathbf{f}$ by minimizing the prediction error on the training set
- **Testing:** apply $\mathbf{f}$ to a never before seen *test example* $\mathbf{x}$ and output the predicted value $y = f(\mathbf{x})$

Learning—A Two-Step Process

- **Model construction:**
  - A training set is used to create the model.
  - The model is represented as classification rules, decision trees, or mathematical formulae

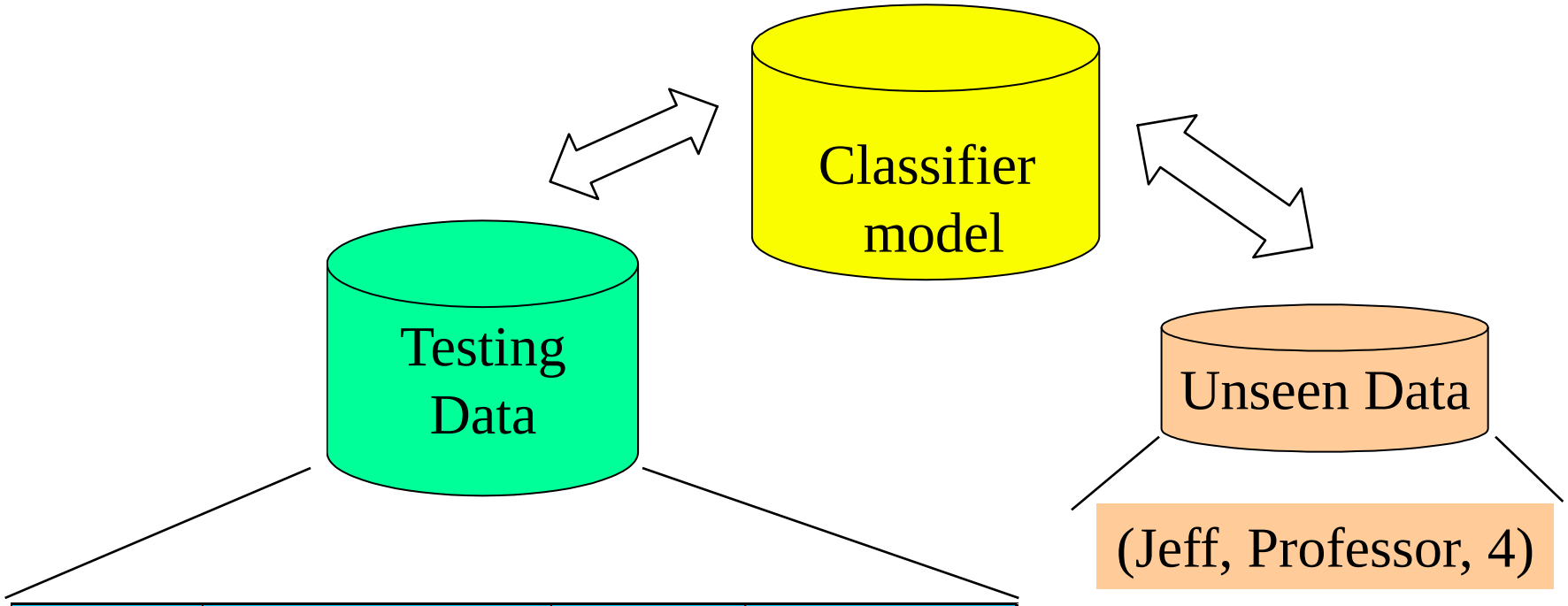
- **Model usage:**
  - the test set is used to see how well it works for classifying future or unknown objects

Step 1: Model Construction



NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

Step 2: Using the Model in Prediction



NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

(Jeff, Professor, 4)

Tenured?

Yes

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Applications of Machine learning

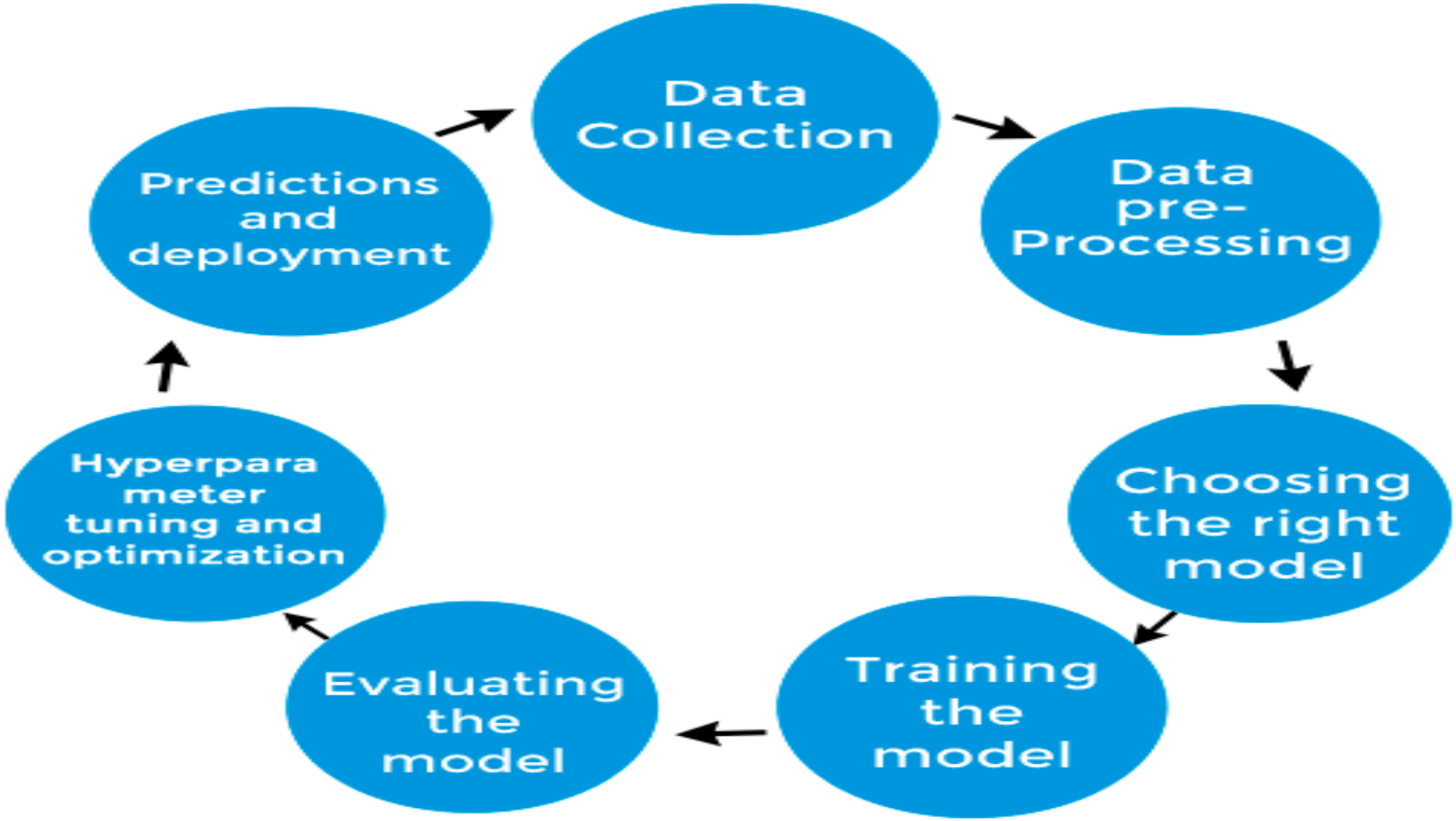
- Prediction (weather, medical, agricultural yield, etc.)
- Face detection; character recognition;
- Surveillance and security system
- Object detection and recognition
- Natural language processing (word sequence predication, spelling and grammar checker, speech recognition (dialog system),translation system, information retrieval, news text classification)
- Image segmentation
- Multimedia event detection
- Economical and commercial usage
- Outlier detection: Exceptions that are not covered by the rule, e.g., fraud
- …… many many more

Challenges in Machine Learning

- Efficiency and scalability of machine learning algorithms
- Handling high-dimensionality
- Handling noise, incomplete and imbalanced data
- Pattern evaluation and knowledge integration
- Protection of security, integrity, and privacy in machine learning
- Data acquisition and representation issues
- Degree of interpretability for predictive power
- Deployment issues

Basic Steps in Machine Learning

1. **Data collection**
   - "training data", **mostly** with "labels" provided by a "teacher";
2. **Data preprocesing**
   - Clean data to have homogenity
3. **Feature engineering**
   - Select represenatative features to improve performance
4. **Modeling**
   - choose the class of models that can describe the data
5. **Estimation/Selection**
   - find the model that best explains the data: simple and fits well;
6. **Validation**
   - evaluate the learned model and compare to solution found using other model classes;
7. **Operation**
   - **Apply learned model to new "test" data or real world instances**


Data Collection
Data pre-Processing
Choosing the right model
Training the model
Evaluating the model
Hyperpara meter tuning and optimization
Predictions and deployment

Machine Learning and Python

Python is the most popular programming language specially to implement machine learning because of its simplicity, vast collection of libraries and easiness.

Python libraries are collection of codes and functions that can be used in a program for a specific task.

➢Some popular Python machine learning libraries are as follows −

- Scikit-learn
- Matplotlib
- NumPy
- Pandas
- SciPy

- **Scikit-learn**, a popular open-source library built on NumPy and SciPy, is used to implement machine learning models and statistical modeling. It supports supervised and unsupervised learning. It provides various tools for implementing data pre-processing, feature selection, model selection, model evaluation, and many other tasks.

- To install use command **pip install scikit-learn**

- **NumPy** is a general purpose array and matrix processing package used for scientific computing and to perform a variety of mathematical operations like linear algebra, Fourier transform and others.

- To import use **import numpy as np**

- To install **Pip install numpy**

- **Pandas** is a powerful library for data manipulation and analysis.

  - To import this library use **import pandas as pd**

  - To install this library use **Pip install pandas**

- **SciPy** is an open-source library that performs scientific computing on large datasets. It is easy to use and fast to execute data visualization and manipulation tasks.

- **Matplotlib** is a popular plotting library usually used for data visualization, to create graphs, plots, histograms and bar charts. It provides tools and functions for data analysis, exploration and presentation tasks.

```
pip install matplotlib
import matplotlib.pyplot as plt
```

ANY QUESTION????